

Laboratorio de Microeconomía I

Centro de Investigación y Docencia Económicas

Maestría en Economía - 2025

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Tarea 6

Exercises

Ex. 1.30 In the two-good case, the level sets of the indirect utility function in price space are sets of the form

$$\{(p_1, p_2) \mid v(p_1, p_2, y) = v^0\} \quad \text{for } v^0 \in \mathbb{R}.$$

These are sometimes called **price-indifference curves**. Sketch a possible map of price-indifference curves. Give separate arguments to support your claims as to their slope, curvature, and the direction of increasing utility.

Ex. 1.31 Show that the indirect utility function in **Example 1.2** is a quasiconvex function of prices and income.

Ex. 1.54 The n -good Cobb–Douglas utility function is

$$u(\mathbf{x}) = A \prod_{i=1}^n x_i^{\alpha_i}$$

where $A > 0$ and $\sum_{i=1}^n \alpha_i = 1$.

(a) Derive the Marshallian demand functions.

(b) Derive the indirect utility function.

Ex. 3.G.15 Consider the utility function

$$u(x_1, x_2) = 2x_1^{1/2} + 4x_2^{1/2}$$

(a) Find the demand functions for goods 1 and 2 as they depend on prices and wealth.

(b) Find the indirect utility function and verify all the properties of an indirect utility function.

Appendix

1.4.1 The Indirect Utility Function

The ordinary utility function $u(\mathbf{x})$ is defined over the consumption set X and represents consumer preferences directly, as we have seen. It is therefore referred to as the **direct utility function**. Given prices $\mathbf{p}$ and income y , the consumer chooses a utility-maximizing bundle $\mathbf{x}(\mathbf{p}, y)$. The level of utility achieved when $\mathbf{x}(\mathbf{p}, y)$ is chosen thus will be the highest level permitted by the consumer's budget constraint facing prices $\mathbf{p}$ and income y . Different prices or incomes, giving different budget constraints, will generally give rise to different choices by the consumer and so to different levels of maximised utility. The relationship among prices, income, and the maximised value of utility can be summarised by a real-valued function $v : \mathbb{R}_+^n \times \mathbb{R}_+ \rightarrow \mathbb{R}$ defined as follows:

$$v(\mathbf{p}, y) = \max_{\mathbf{x} \in \mathbb{R}_+^n} u(\mathbf{x}) \quad \text{s.t.} \quad \mathbf{p} \cdot \mathbf{x} \leq y. \quad (1.12)$$

The function $v(\mathbf{p}, y)$ is called the **indirect utility function**. It is the maximum-value function corresponding to the consumer's utility maximisation problem. When $u(x)$ is continuous, $v(\mathbf{p}, y)$ is well-defined for all $\mathbf{p} \gg 0$ and $y \geq 0$. If $u(\mathbf{x})$ is strictly quasiconcave, the solution is unique and we can write it as $x(\mathbf{p}, y)$, the consumer's demand function. Hence,

$$v(\mathbf{p}, y) = u(\mathbf{x}(\mathbf{p}, y)) \quad (1.13)$$

Theorem 1.6 Properties of the Indirect Utility Function

If $u(\mathbf{x})$ is continuous and strictly increasing on $\mathbb{R}_+^n$, then $v(\mathbf{p}, y)$ defined in (1.12) is

1. Continuous on $\mathbb{R}_{++}^n \times \mathbb{R}_+$,
2. Homogeneous of degree zero in $(\mathbf{p}, y)$,
3. Strictly increasing in y ,
4. Decreasing in $\mathbf{p}$,
5. Quasiconvex in $(\mathbf{p}, y)$.

Moreover, it satisfies

6. Roy's identity: If $v(\mathbf{p}, y)$ is differentiable at $(\mathbf{p}^0, y^0)$ and $\partial v(\mathbf{p}^0, y^0)/\partial y \neq 0$, then

$$x_i(\mathbf{p}^0, y^0) = -\frac{\partial v(\mathbf{p}^0, y^0)/\partial p_i}{\partial v(\mathbf{p}^0, y^0)/\partial y}, \quad i = 1, \dots, n.$$

Example 1.2 In Example 1.1, the direct utility function is the CES form, $u(x_1, x_2) = (x_1^\rho + x_2^\rho)^{1/\rho}$, where $0 \neq \rho < 1$. There we found the Marshallian demands:

$$x_1(\mathbf{p}, y) = \frac{p_1^{r-1}y}{p_1^r + p_2^r}, \quad x_2(\mathbf{p}, y) = \frac{p_2^{r-1}y}{p_1^r + p_2^r}, \quad (\text{E.1})$$

for $r \equiv \rho/(\rho - 1)$. By (1.13), we can form the indirect utility function by substituting these back into the direct utility function. Doing that and rearranging, we obtain

$$\begin{aligned} v(\mathbf{p}, y) &= \left[(x_1(\mathbf{p}, y))^\rho + (x_2(\mathbf{p}, y))^\rho \right]^{1/\rho} \\ &= \left[\left(\frac{p_1^{r-1}y}{p_1^r + p_2^r} \right)^\rho + \left(\frac{p_2^{r-1}y}{p_1^r + p_2^r} \right)^\rho \right]^{1/\rho} \\ &= y \left[\frac{p_1^r + p_2^r}{(p_1^r + p_2^r)^\rho} \right]^{1/\rho} \\ &= y(p_1^r + p_2^r)^{-1/r} \end{aligned} \quad (\text{E.2})$$